Higher National Diploma in Engineering (Civil)

2nd year, First Semester Examination - 2021

CE2110 - Water and Environmental Engineering

Q1.

a) Define the term "air pollution" and mention four sources of air pollution. (05 marks)
b) Water pollution is caused due to industrial waste released into lakes, ponds and other water bodies. Briefly discuss this matter with the examples. (05 marks)
c) Briefly explain about global warming and write down five (5) consequences can be happened in the earth due to the global warming. (06 marks)
d)

i. Define the terms "Point source pollution" and "Non-point source pollution". (04 marks)

ii. "Non-point source pollution effect is high than the effect from point source pollution". Briefly explain about this statement by giving examples. (05 marks)

(Total 25 marks)

Q2.

a) Define the following terms/words23. (05 marks) 24I. Sewer 25

II. Sewerage

III. Dry weather flow

IV. Infiltration

V. Inflow

b) Draw the flow rate vs. time graph to show the variation in sewage flow of a community. (02 marks)

c) Mention four purposes of the manholes33. (04 marks)

d) What is the meaning of "sanitary sewer overflow" and briefly explain three risks which is associated with it. (05 marks)

e)A city located in an urban area, serving a population of 60,000 spread over an area of hectares. Find the design discharge for the combined sewer line by assuming rate of water

supply of 180 lcpd and out of this total supply only 70% reaches in sewer as wastewater. In this area, it is expected to rainfall of 30mm in 24 hrs and runoff coefficient is 0.4.

(Clearly state your assumptions)

- Sewer length: **5.6 km**
- Infiltration rate: $75m^3$/km. s

(9 marks)

(Total 25 marks)

Q3.

a) Define the term "self-cleansing velocity" and briefly explain the way of maintain that in the sewer mains. (04 marks)

b)

i. The maximum and the minimum velocities are specified in the sewer lines. Give the reasons for that. (04 marks)

ii. Clearly state the maximum and the minimum velocities for sanitary sewer mains. (02 marks)

c) Mention three (03) reasons for clogging of the sewers and give the prevention methods of those. (06 marks)

d) A trunk sewer made of concrete (rougliness co-efficient, n=0.012) with a diameter of 350mm, carries $0.0113m^3$/Sec of discharge when half full (1/2). Find the:

i. Depth of flow

ii. Velocity

iii. Slope of the pipeline

(Use the graph 1) (09 marks)

(Total 25 marks)

Q4.

a) Write down four (04) factors should be considered regarding the site condition before design of the sewer network. (04 marks)

b) Write down four types of materials that can be used for the sewer construction. (04 arks)

c) Mention three important details should be included in the design profile (Longitudinal section) of the sewer line. (03 marks)

d) Briefly discuss three major construction steps in sewer construction works with suitable sketches. (09 marks)

e) Briefly explain about the Water test (sewer testing) done for newly constructed sewer mains. (05 marks)

(Total 25 marks)

Q5.

a) Draw the typical flow-diagram for preliminary wastewater treatment stages. (04 marks)

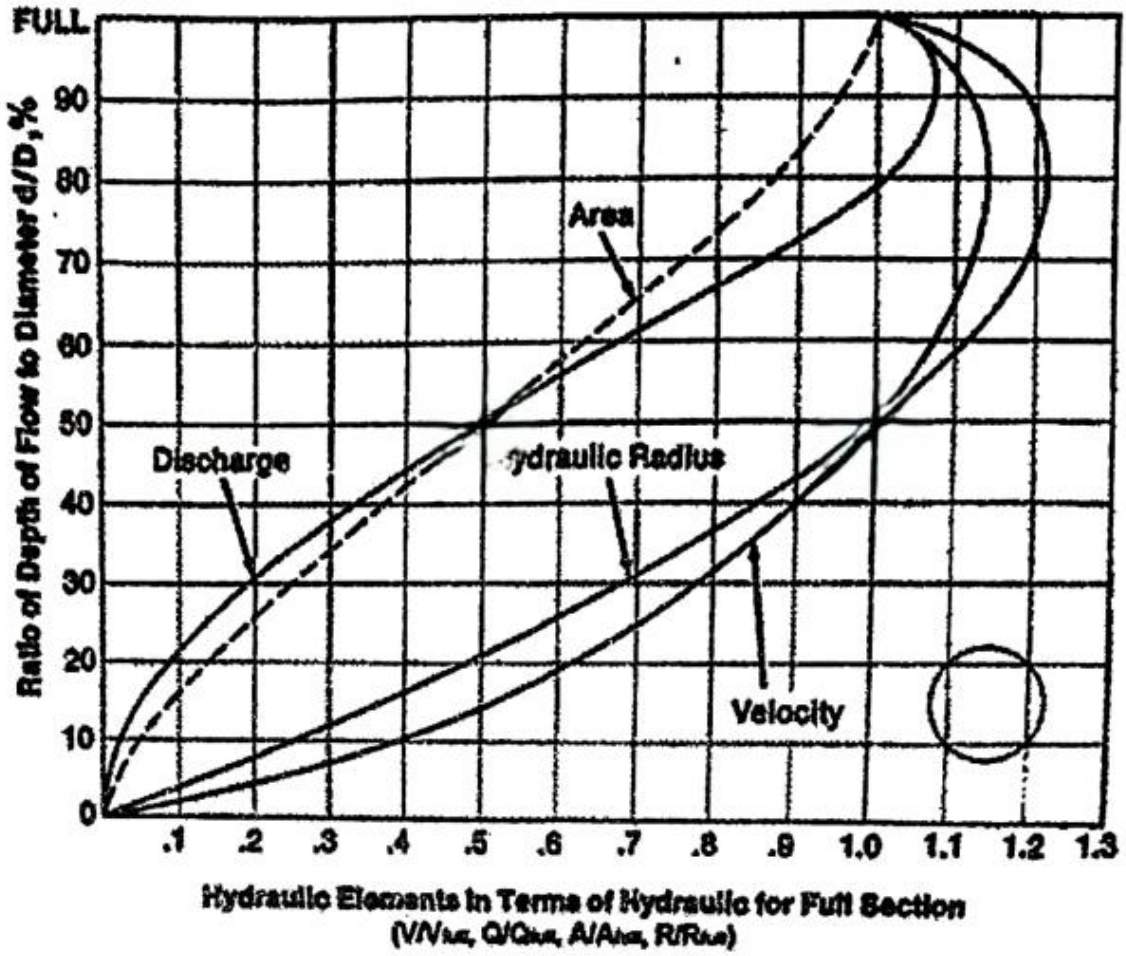
b) Mention the purpose of screening at the inlet of wastewater treatment plant. (03 marks)

c) Briefly explain about the eutrophication (mention the reasons and consequences of it). (05 marks)

d) Wastewater characteristics can be divided into three broad categories. Briefly explain these three categories with two examples for each. (09 marks)

e) Determine Ultimate BOD for a wastewater whose 5-day 20°C BOD is 200 mg/l. The reaction constant K is 0.23d^{-1}. (04 marks)

(Total 25 marks)


FULL
90
80
70
60
50
40
30
20
10
0
Ratio of Depth of Flow to Diameter d/D, %
Area
Discharge
ydraulic Radius
Velocity
.1
.2
.3
.4
.5
.6
.7
.8
.9
1.0
1.1
1.2
1.3
Hydraulic Elements in Terms of Hydraulic for Full Section

Graph 1